Pokhara University
Faculty of Science and Technology

Course No.: CMP 517 (3 Credits)

Full marks: 100

Course title: **Research Techniques**

Pass marks: 60

Nature of the course: Theory

Total Lectures: 45 hrs

Level: Master

Program: ME/MSc (Computer)

1. Course Description

This course is designed to provide the students with the skills and methodologies required for conducting high-quality research in computer science and computer engineering. Covering both theoretical and practical aspects, it includes research design, literature review, problem formulation, data collection, and presentation techniques. Students will learn to critically evaluate research papers, design experiments, analyze data, and write technical papers. Emphasis is placed on applying these skills to solve complex problems and contribute to the body of knowledge in computer engineering.

2. General Objectives

- Develop a clear understanding of research methodologies and techniques.
- Acquaint the student with the knowledge to identify and formulate research problems in computer engineering.
- Provide the students with skills of literature review, experimental design, and data analysis.
- Acquaint the students with the proficiency in technical writing, publishing, and presenting research findings.
- Provide the knowledge of ethical considerations in research.

3. Methods of Instruction

Lecture, Discussion, Readings, Case Studies, Project and Research Works.

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4. Contents in Detail

Specific Objectives	Contents
• Organize a scientific research process to solve a real world problem.	**1. Introduction to Research (5 hrs)** 1.1. Research: Objectives and Characteristics 1.2. Types of Research 1.3. The 6 P's of Research 1.4. Research Process 1.5. Research in Computer Science and Engineering
• Apply literature review and identify a research gap to formulate a problem.	**2. Literature Review (6 hrs)** 2.1. Purpose of Literature Review 2.2. Literature Resources and Techniques 2.3. Conducting a literature review 2.4. Referencing and Citation 2.5. Evaluating literature review 2.6. Identifying problem: Research question / problem statements and hypothesis 2.7. Case Study: Prepare a review paper.
• Apply the research design and strategies based on the nature of the problem chosen to solve.	**3. Research Methodology (8 hrs.)** 3.1. Components of Research Methodology: Research problem, Objectives, Research Design, Data Collection, Sampling Techniques, Data Analysis, Hypothesis Formulation, Ethical Consideration and Validation and Reliability 3.2. Research Design: Experimental, Exploratory, Descriptive and Explanatory 3.3. Research Strategies: Experimental Research, Case Study, Action Research, Theoretical Research, Simulation-based Research. 3.4. Multidisciplinary Research Approaches
• Use appropriate data collect methods to generate the data required for the chosen problem and evaluate and validate the data.	**4. Data Collection (6 hrs.)** 4.1. Primary and Secondary Data 4.2. Data Collection Methods 4.2.1. Surveys 4.2.2. Experiments 4.2.3. Questionnaires 4.2.4. Interviews 4.2.5. Observation 4.2.6. Experiments 4.2.7. Documents 4.3. Data Preparation 4.4. Data Sampling and Its Techniques 4.5. Evaluation and Validation of Data

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• Apply the data analysis and visualization tools techniques to analyze and visualize the pattern in the data.	**5. Data Analysis (12 hr)** 5.1. Quantitative Data Analysis 5.1.1. Types of Quantitative Data 5.1.2. Data Coding 5.1.3. Data Presentation: Tables, Bar charts, Scatter Graph, Line graph 5.1.4. Analysis Techniques: Descriptive Statistics, Inferential Statistics (Z-test, t-test, Chi-square test, ANOVA), Regression Analysis, Time Series Analysis and Multivariate Analysis 5.2. Qualitative Data Analysis 5.2.1. Qualitative Data 5.2.2. Analyzing Textual Data 5.2.3. Analyzing Non-Textual Data 5.3. A detailed Case Study in data visualization and analysis using Python libraries: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-learn
• Apply the research ethics during the entire research work.	**6. Ethical Issues in Research (4 hrs)** 6.1. Right of People directly involved 6.2. Responsibilities of an Ethical Researcher 6.3. Data Privacy and Security 6.4. Plagiarism and Intellectual Property 6.5. Algorithmic Bias and Fairness 6.6. Environmental Responsibility 6.7. Conflict of Interest 6.8. Copyright and Patent 6.9. ACM Code of Ethics and IEEE Code of Ethics
• Write the research work and present it in research papers, thesis, conference and journal.	**7. Presentation of the Research (4 hrs)** 7.1. Writing up the research 7.2. Written Presentation 7.2.1. Academic papers 7.2.2. Theses and dissertations 7.2.3. Technical reports 7.3. Visual Presentations 7.3.1. Posters 7.3.2. Infographics 7.3.3. Demonstration 7.4. Oral Presentation 7.4.1. Conference talks 7.4.2. Thesis defense 7.5. Conferences and Journal [Students must present (oral) a review paper or any research paper prepared by them in class or national/international conference]

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5. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, internal assessment, project and research works etc. The internal evaluation scheme for this course is as follows (but not limited to):

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Attendance & Class Participation	10%	60	Semester-End examination	40
Assignments	20%			
Internal Assessment	40%			
Research/Project/Case Study Work	30%			
Total Internal		60	**Total External**	40
Full Marks: 60 + 40 = 100				

Student Responsibilities

Each student must secure at least 60% marks separately in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Text Books and Reference Books

Text Book

1. Oates, B. J., Griffiths, M., & McLean, R. (2022). *Researching information systems and computing*. Sage..

Reference Books

1. Creswell, J. W., & Creswell, J. D. (2017). *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage publications.
2. Flick, U. (2022). *An introduction to qualitative research*. Sage Publications.
3. Kothari, C. R., (2009). *Research Methodology: Methods and techniques*. New International Publisher

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